

OTHER COMMON DENTAL MALOCCLUSIONS AND THERE PRINCIPLE OF MANAGMENT

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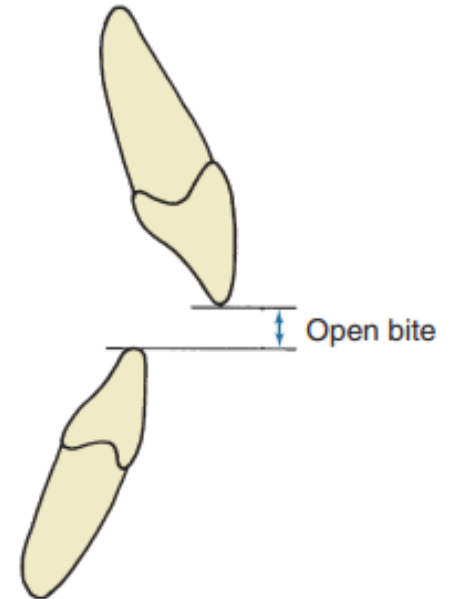
LEARNING OBJECTIVES

- ✓ *EXPLAIN PRINCIPLES OF MANAGMENTS OF OPEN BITE,DEEP BITE AND CROSS BITE*

PRINCIPLES OF MANAGEMENT OF OPEN BITE

OPEN BITE

- An open bite is said to exist when there is a **lack of vertical overlap** between the maxillary and mandibular teeth.
- In normal circumstances the mandibular dental arch is contained within the maxillary arch.
- In other words the maxillary teeth overlap the mandibular teeth labially and buccally.
- Open bites can exist in the ***anterior as well as the posterior region***.
- An open bite present in the anterior segment is the most unaesthetic, as the patient has to bring his tongue anteriorly between the teeth and the lips during speech and while swallowing.
- Posterior open bites may hamper mastication and are more difficult to treat.



CLASSIFICATION OF OPEN BITE

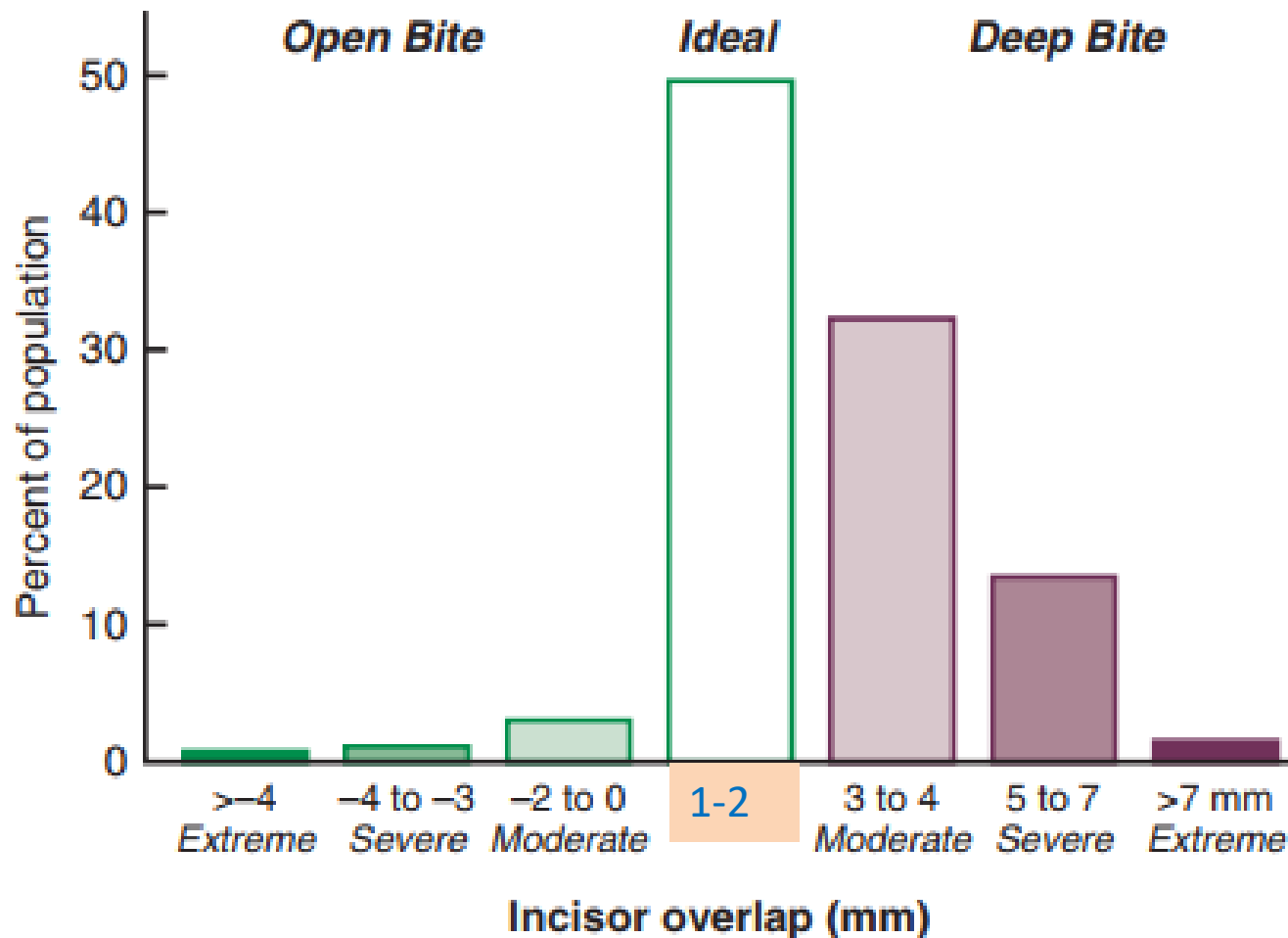
I. Based on the location of the open bite, they may be classified as:

- Anterior open bite(single tooth, segmental)
- Posterior open bite(unilateral, bilateral)

II. Based on the dental or skeletal components involved(based on source), open bites can be classified as:

- Skeletal open bite, or
- Dental open bite.

Degree of severity of Open bite



ETIOLOGY OF ANTERIOR OPEN BITE

- ✓ Habits
 - Anterior tongue thrust
 - Digit sucking habits
 - Mouth breathing
- ✓ Abnormally increased tongue size
- ✓ Inherited or acquired, abnormal growth pattern

ANTERIOR OPEN BITE

- Anterior open bite is the most commonly encountered severity of open bite.
- ***Majority of the anterior open bites encountered in day-to-day clinical practice are dental in nature .***
- ***They are usually associated with a local cause, which has to be removed for the correction of the malocclusion .***
- The persistence of pernicious habit can lead to the malocclusion acquiring a skeletal component or it could be the result of a hereditary skeletal pattern .
- Skeletal anterior open bite can occur if there is incoherent growth of the maxilla and/or mandible and/or anterior cranial base.
- ***Is of two types: DENTAL OR SKELETAL***

Features of dental anterior open bites

➤ Intraoral features:

- 1 Open bite limited to the anterior segment, often asymmetrical.
2. Proclined maxillary and/or mandibular incisors.
3. Spacing between maxillary and/or mandibular anteriors.
4. Narrow maxillary arch is a possibility.
5. “Fish mouth” appearance.

➤ Extraoral features:

- No unusual features.



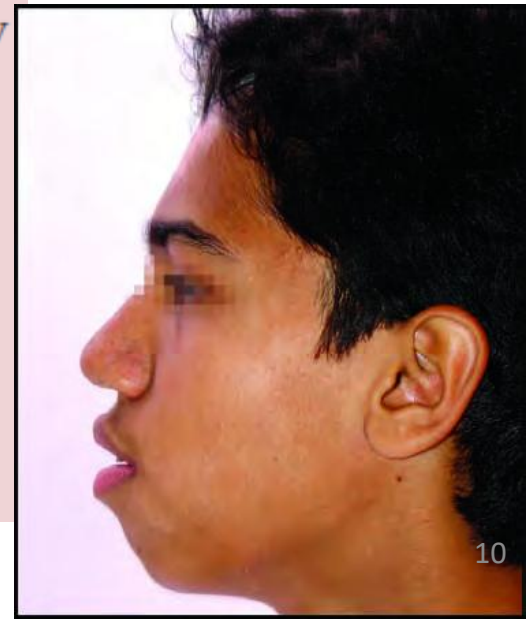
Features of skeletal anterior open bites

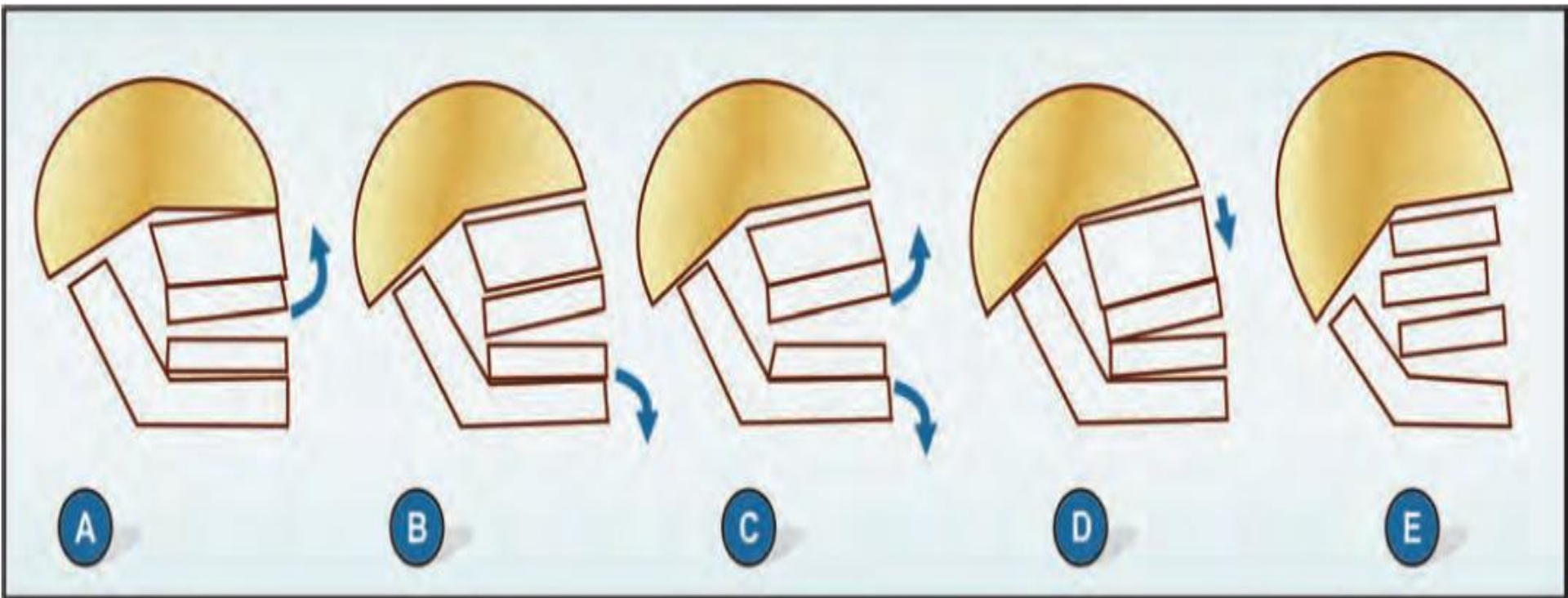
Extraoral features:

1. Long face due to increased lower anterior face height
2. Incompetent lips
3. An increased mandibular plane angle
4. An increased gonial angle
5. Marked antegonial notch
6. A short mandible is a possibility
7. Maxillary base may be more inferiorly placed (vertical maxillary excess)
8. The angle formed by the mandibular and maxillary planes is also increased

Intraoral features:

1. Mild crowding with upright incisors
2. Gingival hypertrophy
3. Maxillary, occlusal and palatal planes tilt upwards
4. Mandibular occlusal plane canted downwards





Figs 54.2A to E: Skeletal anterior open bite (A) Due to upward maxillary rotation, (B) Due to downward mandibular rotation, (C) Due to combination of downward rotation of mandible and upward rotation of maxilla, (D) Due to vertical maxillary excess, (E) Due to an increased flexure angle

MANAGEMENT OF ANTERIOR OPEN BITE (dental)

- The appliances used for the corrections of anterior open bites are usually used in conjunction with the **habit breaking** appliances used for the elevation of the underlying etiologic cause.
- Unless the treatment of the underlying etiologic factor is delayed and the patient is seen as an adolescent or an adult, anterior open bites have a tendency to regress spontaneously with the **removal of the underlying cause** .
- In cases with a minor skeletal component or where the correction is not seen spontaneously, **fixed appliances should be used in conjunction with a removable or fixed habit-breaking appliance** .
- Box elastics of medium to heavy forces may be used for the correction of mild to moderate open bites



a/ Spontaneous correction of a mild anterior open bite with the wearing of a removable habit breaking appliance



b/ Fixed appliances used along with a fixed tongue crib for the correction of anterior open bite



c/ Moderates force box elastics used to close the anterior open bit along with fixed appliances

skeletal anterior open bite

- A chin cup with a vertical pull head cap may be used for the correction of anterior open bites in the ***pre-adolescent age*** group. (***GROWTH MODIFICATION***)
- Skeletal open bites in adults should be treated ***surgically*** after the correction of the existing habit.
- Surgery generally involves the Le-Fort I osteotomy to impact the maxilla posteriorly.
- Muscle-retraining exercises may be required following the surgical correction.
- Mild to moderate case-level the occlusal curve with extraction and retraction(***Camouflage***)



a/ Pre-adolescent patient with skeletal anterior open bite



b/ Patient wearing the chin cup with a vertical pull head cap

POSTERIOR OPEN BITE

- Posterior open bites are characterized by a lack of contact between the posterior teeth when the teeth are brought in occlusion.
- Posterior open bites are relatively rare and are caused mainly because of a lateral tongue thrust habit or submerged/ankylosed posterior teeth.

CORRECTION OF POSTERIOR OPEN BITE

- Since lateral tongue thrust is the most frequently encountered etiologic factor, the use of lateral tongue spikes either fixed or incorporated in a removable appliance, form the first line of treatment.
- *Vertical elastics used along with fixed orthodontic appliances can be used once the lateral tongue thrust habit has been controlled.*
- Fixed appliances are the most frequently used means for the correction of submerged and impacted teeth.



a/ Posterior open bite caused by a lateral tongue thrust habit



b/ Lateral tongue spicker incorporated in an acrylic appliance

PRINCIPLES OF MANAGEMENT OF DEEP BITE

DEEP BITE

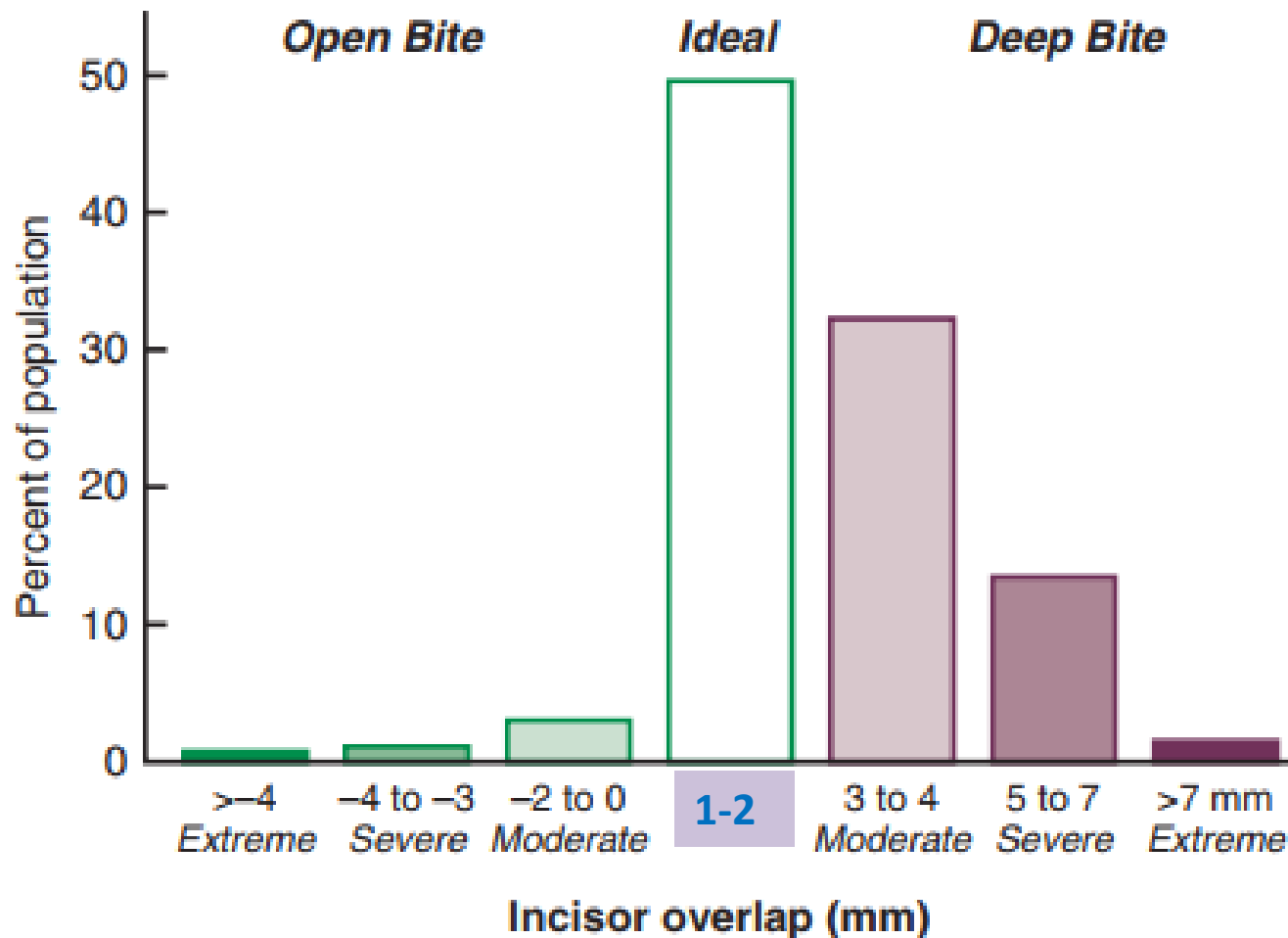
- Deep bite is one of the frequently seen malocclusions next to crowding.
- It is said to be one of the most perpetuating and damaging malocclusions .
- It may jeopardize the periodontal support, occlusion itself or TMJ .
- *The excessive overbite is a complex orthodontic problem that may involve a group of teeth or whole dentition, alveolar bone, of maxillary and mandibular basal bones, and/or soft tissue of the face.*
- The management of this problem demands a careful diagnostic analysis, treatment plan, and selection of appropriate treatment therapy



- The term "**overbite**" applies to the distance which the maxillary incisal margin closes vertically past the mandibular incisal margin .
- Normally the lower incisal edges contact the lingual surface of the upper incisors at or slightly above the cingulum (i.e., normally there is 1 to 2 mm overbite).

- **Grabber** has defined 'Deep bite' as a condition of excessive overbite, where the vertical measurement between the maxillary and mandibular incisal margins is excessive when the mandible is brought into habitual or centric occlusion'.

Degree of severity of Deep bite



Classification

1. According to its origin;

- a) Dental deep bites (Simple).
- b) Skeletal deep bite (Complex).

2. According to functional classification;

- a) True deep bite.
- b) Pseudo deep bite.

3. Depending on the extent of deep bite

- incomplete over bite
- complete over bite

4. According to dentition;

- a) Primary dentition deep bite.
- b) Mixed dentition deep bite.
- c) Permanent dentition deep bite



Fig. 7 Skeletal deep bite associated with Class II div 2 malocclusions: the face is muscular, straight to mild convex. Reduced anterior facial height, particularly the lower facial height. The mentolabial sulcus is deep. Intraorally the maxillary and mandibular incisors are retroclined and there is near 100% vertical overlap



Fig. 8 Deep bite due to vertical maxillary excess and downward and forward (clockwise) rotation of maxilla

a. Simple (dental) deep bite

- A simple deep bite is localized to the teeth and alveolar processes.
- ***Dental deep bites occur due to over-eruption of anteriors or infraocclusion of molars.***
- The result may be labial version of the upper incisors and impingement of the lowers into the palatal mucosa
- A majority of the problems in this category are created by the loss of permanent teeth causing a lingual collapse of maxillary or mandibular anterior teeth.
- This kind of deep bite is characterized by the absence of any ***skeletal complicating*** features which are seen in skeletal deep bites
- ***In the mandibular dentition, it may manifest as a deep curve of Spee or a reverse curve of Spee in the maxillary dentition.***
- ***These patients frequently show temporomandibular dysfunction and a limited range of functional occlusal movements.***



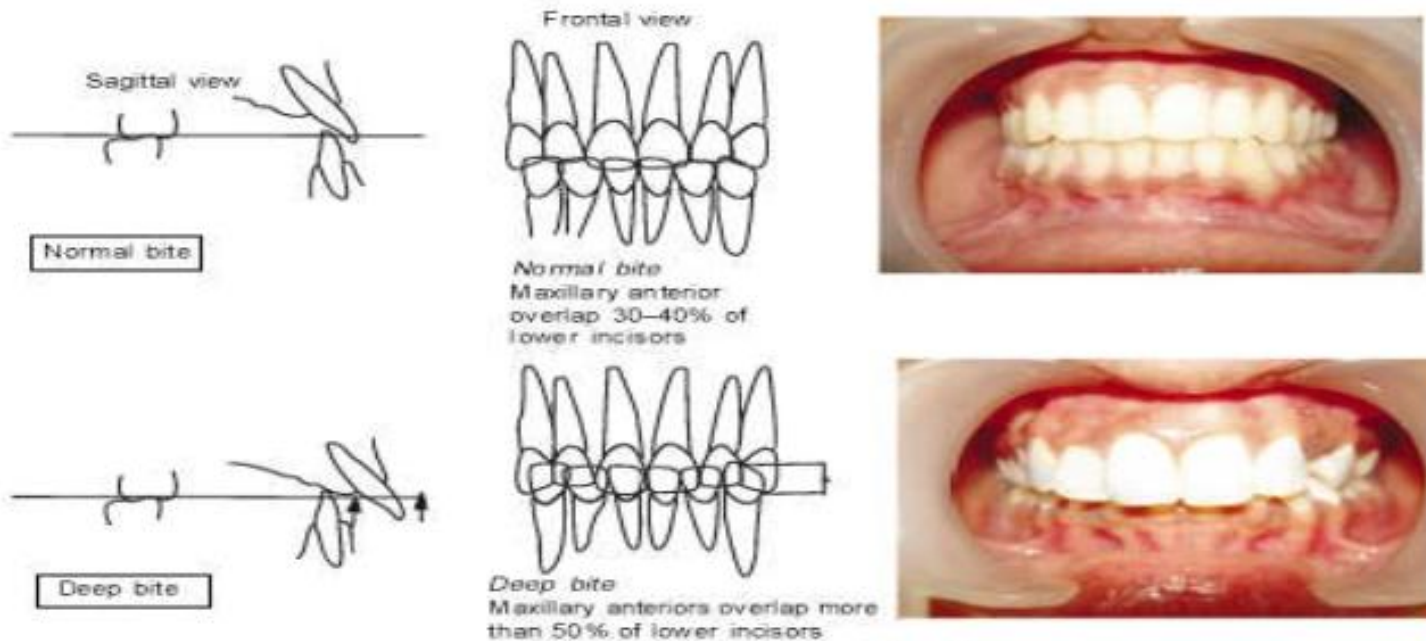
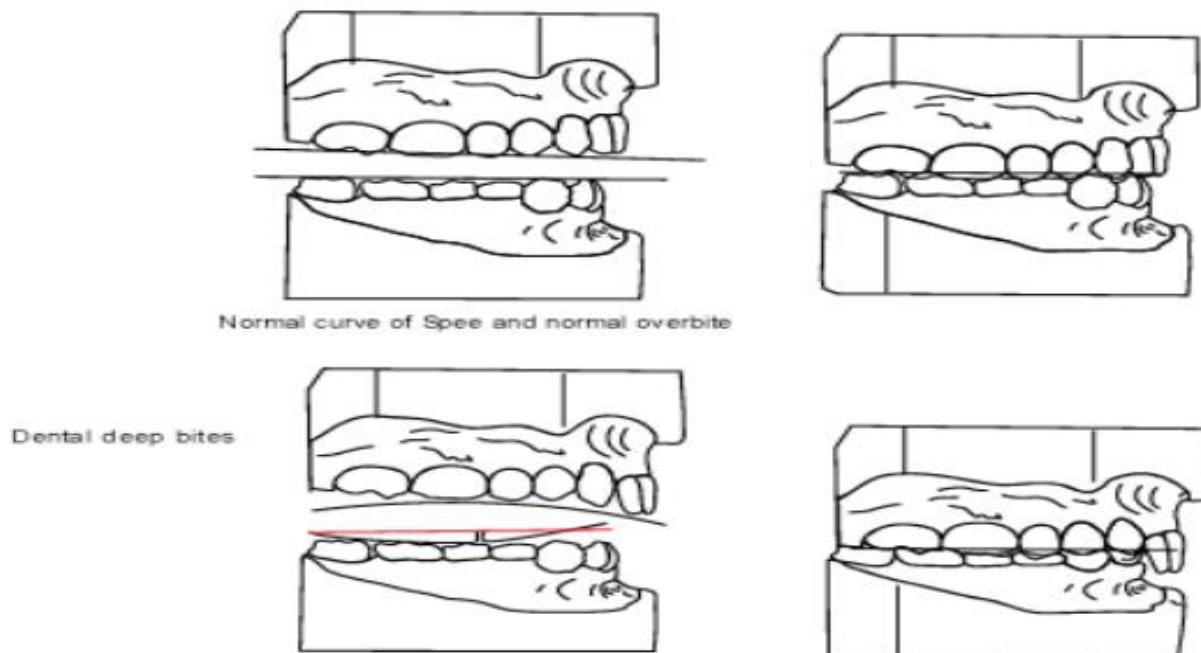


Fig. 1 Normal bite and deep bite

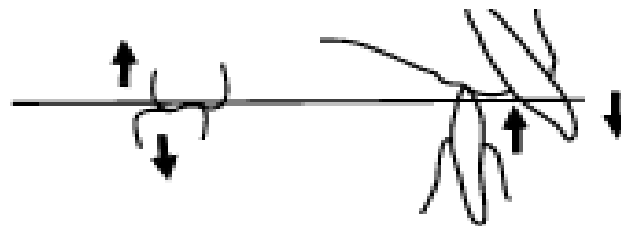


Exaggerated curve of Spee in lower arch and exaggerated curve of Spee in maxillary arch. This is due to overeruption of anteriors or undereruption of posteriors

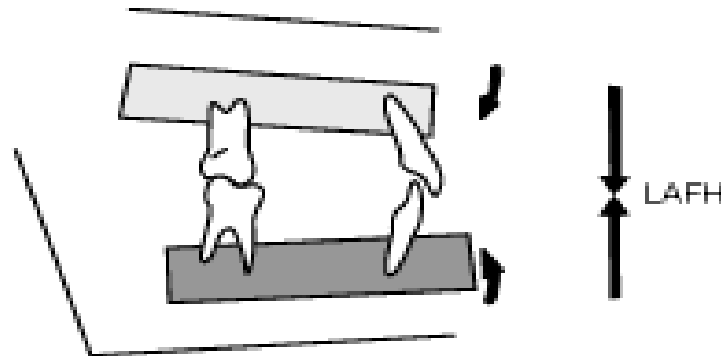
Fig. 2 Normal bite and deep bite

b. Complex (skeletal) deep bite

- Complex deep bite is a deep bite associated with basic skeletal features with which the alveolar process cannot cope.
- A skeletal type of overbite may be due either to malrelationship of alveolar bones and/or underlying mandibular or maxillary bones or to an overgrowth or undergrowth of one or more alveolar segments
- The diminished anterior vertical height of the face is also an important criterion for diagnosis of skeletal deep overbites.



Dental deep bite results from overeruption of anterior teeth or undereruption of molars. True deep bite results from infraocclusion of posteriors. Whereas pseudo-deep bite results from supraeruption of posteriors



Skeletal deep bite results with jaw bases moving towards each other (convergent jaw bases). The lower anterior facial height (LAFH) is diminished

Fig. 3 Dental and skeletal deep bites



In low-angle cases, the two planes meet beyond the occipital region. In severe low-angle cases the two planes run almost parallel to each other. Low FMA angle cases results from forward or anticlockwise rotation of the mandible. In this cases lower facial height is decreased with tendency towards deep bite

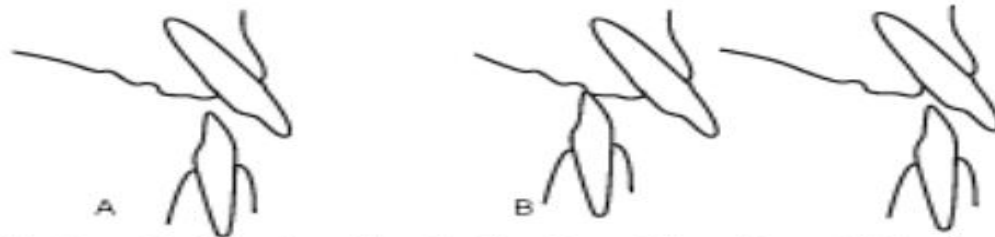
True and pseudo-deep overbite

True deep overbite	Pseudo-deep overbite
This is caused by infraocclusion of the posterior segments ie..molars	is caused by overeruption of the anterior teeth that already has normal eruption of the posterior segment teeth
Seen in class II div II	Seen in class II div I malocclusions
It is often the result of a lateral tongue posture of tongue thrust . The interposition of tongue prevents the eruption of the posterior teeth. It can also occur due to premature loss of posterior teeth	It is the result of overeruption of the incisors. Due to the presence of the increased overjet, the lower incisors to over-erupt until they meet the palatal mucosa.
These patients have near flat curve of spee.	These patients hence exhibit an excessive curve of Spee
There is a large interocclusal clearance	The inter-occlusal clearance is usually normal or small as the molars are fully erupted.

Some Class II, division II, malocclusion with adequate lip line relationships are good examples	Some Class II division I, malocclusions with a "gummy" smile and a poor lip line relation can fall into this category
Treatment in the mixed dentition period requires the elimination of environmental factors that are inhibiting eruption of the posterior teeth. Ideal for functional appliance therapy	Incisors cannot be intruded effectively using functional methods during mixed dentition .
Extrusive mechanics of molars possible	All possible intrusive mechanics on the incisor teeth with fixed appliances is usually indicated . extrusion of molars is possible only to a limited extent

3. Incomplete and complete deep bite

- Incomplete over bite is an incisor relationship in which the lower incisors fail to occlude with either the upper incisors or the mucous of the palate when the teeth are occluded
- Complete over bite on the other hand is a relationship in which the lower incisors contact the palatal surface of the upper incisors or the palatal tissue when the teeth are in centric occlusion .
- This kind of deep bite often results in trauma of the mucous palatal to the maxillary incisors

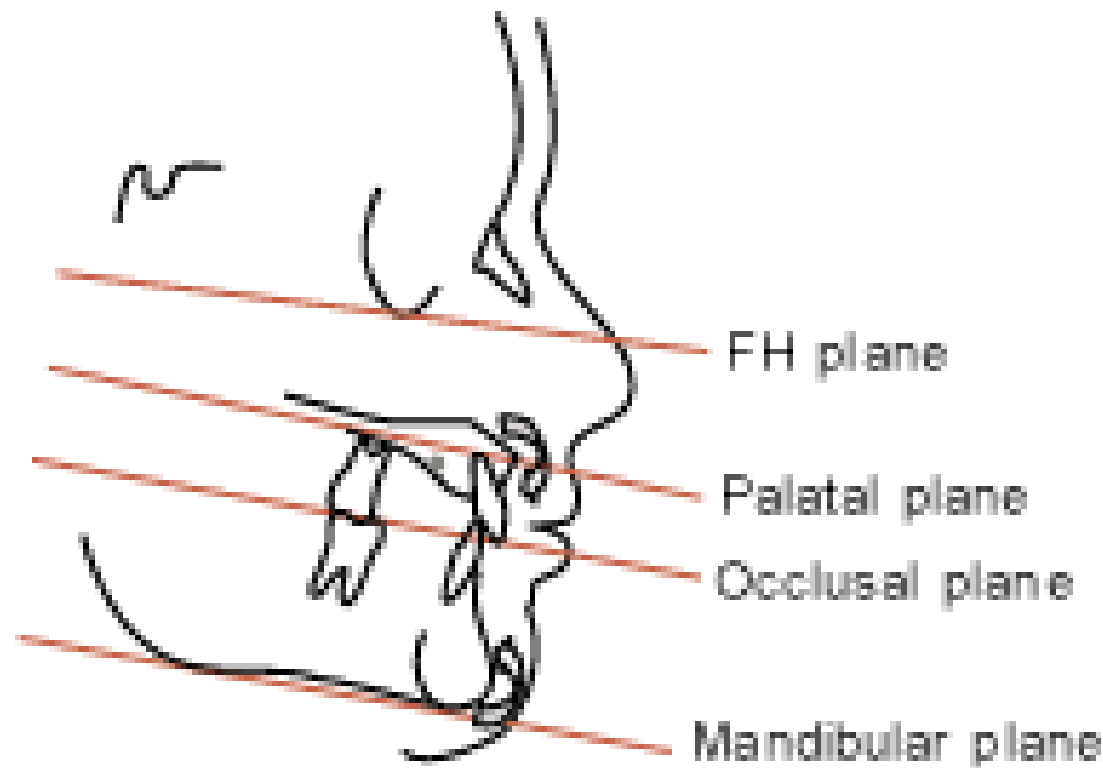


A. Incomplete deep bite—lower incisors do not touch either the palate or the palatal surface of upper incisors
B. Complete deep bite is characterised by lower incisors touching the palate or palatal surface of the upper incisors

Fig. 5 Complete and incomplete bite

Etiology of deep bite

1. **Hereditary** and may follow a genetic pattern or familial condition.
2. **Skeletal**
 - a. An overgrowth or undergrowth of one or more alveolar segments.
 - b. An excess of growth of the ramus and posterior cranial base permits the mandible to rotate upward. Thus Long ramus and short body with decreased gonial angle is characteristic feature
 - c. Convergent upper and lower jaw bases
 - d. Horizontal growth pattern or forward rotation or anticlockwise rotation of the of the lower jaw
 - e. The four planes of the face (intraorbital (FH Plane), palatal, occlusal, and mandibular) as seen from lateral roentgenograms are horizontal and nearly parallel to each other.



Skeletal deep bites are characterised by convergent jaw bases. The maxillary basal bone rotates downwards, i.e., clockwise rotation. The mandible rotates forwards and upwards i.e., anticlockwise rotation. All the four planes of face are horizontal and near parallel to each other. The lower facial height is decreased

Fig. .6 Cephalometric findings in deep bite

3. Dental

- a. Loss and/or mesial tipping of posterior teeth. In other words diminished posterior dental height
- b. Early loss of teeth and lingual collapse of the anterior teeth
- c. Overeruption of the incisor teeth, infraocclusion of the buccal segment or a combination of both.
- d. Overbite may be caused or accentuated by an aberration in the tooth morphology.
- e. Periodontal disease. Bite may deepen if the posterior tooth drift mesially during the pathological migration and worsen the existing condition

4. **Muscular**

- The posterior vertical chain of muscles (masseter, internal pterygoid, temporal) is strong and attached anteriorly on the mandible and stretches in nearly a straight line vertically.
- The molars are directly under the impact of the masticatory forces of this chain.
- When the posterior vertical chain of muscles is strong and anteriorly positioned, a greater depressive action is transmitted to the dentition

5. **Habits**

- a. lateral Tongue thrust swallow
- b. Lip sucking

Diagnosis

- The routine diagnostic aids such as clinical examination, study models and lateral cephalogram are used of the diagnostic exercise .
- The factors contributing to excessive overbite vary with the type of occlusion and skeletal pattern.
- The primary diagnostic problem in both deep bite and open bite is to ascertain the site of the dysplasias whether dental or skeletal.
- The skeletal bite can be differentiated from dental deep bite by cephalometric analysis.
- Postural position is also used in the differential diagnosis of deep bite cases: the freeway space will be larger than normal in cases with inadequate vertical development of the buccal segments and normal in cases of over-eruption of the incisor teeth

Management of deep overbite

- **For Growing patients**

- o Intrude anteriors
- o Erupt posteriors
- o Combination of posterior eruption and anterior intrusion

- **For Non growing patients (little or no growth expected)**

- o Orthognathic surgery
- o Intrusion of anteriors (posterior extrusion invariably relapses)

Factors to be considered before intrusion or extrusion

- Interlabial gap
- Growth pattern whether vertical or Horizontal
- Presence of adequate free way space or interocclusal clearance

PRINCIPLES OF MANAGEMENT OF CROSS BITE

CROSS BITE

- Cross bites are a deviation of the normal bucco-lingual relationship of the teeth of one arch with those of the opposing arch.
- *Graber defined cross bites as a condition where one or more teeth may be malposed abnormally, buccally or lingually or labially with reference to the opposing tooth or teeth.*
- Under normal circumstances the maxillary arch overlaps the mandibular arch both labially and buccally.
- But when the mandibular teeth, single tooth or a segment of teeth, overlap the opposing maxillary teeth labially or buccally, depending upon their location in the arch, a cross bite is said to exist.



CLASSIFICATION OF CROSS BITES

- ❑ Cross bites can be classified according to their location in the arch as
 - Anterior cross bites and
 - Posterior cross bites.
- ❑ Anterior cross bites can be further classified according to the number of teeth involved as
 - Single tooth cross bite , or
 - Segmental cross bite.
- ❑ Posterior cross bites can also be further classified according to the number of teeth involved as
 - Single tooth cross bite, and
 - Segmental cross bite .



Single tooth anterior cross bite



Segmental anterior cross bite



Single tooth posterior cross bite
(maxillary right first molar)



Segmental posterior cross bite

❑ Posterior cross bites may be further classified according to the existence of the cross bite on one side or both the sides of the arch as

- Unilateral cross bite, and
- Bilateral cross bite .



Unilateral cross bite



Bilateral posterior cross bite

Posterior cross bites can also be classified according to the extent of the cross bite as

- *Simple posterior cross bite: This type of cross bite is seen most frequently in clinical practice. Here the buccal cusps of one or more posterior teeth occlude lingual to the buccal cusps of the mandibular teeth*
- *Buccal non-occlusion: Here the maxillary teeth palatal cusp of the occlusion and are placed buccal to the buccal cusp of the mandibular posterior teeth. The condition is also known as scissors bite*
- *Lingual non-occlusion: Here the maxillary posterior tooth or teeth are placed completely palatal to the lingual aspect of the mandibular posterior teeth, i.e. the buccal cusp of the maxillary tooth is palatal/ lingual to the lingual cusp of the mandibular posterior teeth*



Simple posterior cross bite



Scissors bite (maxillary left first premolar)



Lingual/palatal non-occlusion (maxillary left second premolar)

❑ Based on the location of the etiologic factors the cross bites can be classified as

- Dental cross bite
- Skeletal cross bite
- Functional cross bite.

➤ Etiology of dental cross bites



1. Anomalies of number:
 - i. Supernumerary teeth
 - ii. Missing teeth
2. Anomalies of tooth size
3. Anomalies of tooth shape
4. Premature loss of deciduous and / or permanent teeth
5. Prolonged retention of deciduous teeth
6. Delayed eruption of permanent teeth
7. Abnormal eruptive path
8. Ankylosis

➤ Etiology of skeletal cross bites

- They are capable of causing appreciable damage to a person's health and personality as the appearance may be compromised to a larger extent.
- Hereditary (Class III skeletal structure).
- Congenital (cleft lip and palate).
- Trauma at birth (forcep injury causing ankylosis of the TMJ).
- Trauma during growth (ankylosis of the TMJ of retardation of growth in the traumatized bone).
- Trauma after completion of growth (malunion of fracture segments).
- Habits (if not corrected during growth can cause).

Functional cross bites

- *These cross bites are usually* caused due to the presence of occlusal interferences during the act of bringing the jaws into occlusion.
- These can be caused by the early loss of deciduous teeth, decayed teeth or ectopically erupting teeth.
- If not corrected early, these can ultimately lead to skeletal cross bites.

1.CORRECTION OF ANTERIOR CROSS BITE IN THE PREADOLESCENT AGE GROUP

- **Use of Tongue Blade**
- If a cross bite is seen at the time the permanent teeth are making an appearance in the oral cavity, a simple appliance like a tongue blade can correct the developing cross bite.

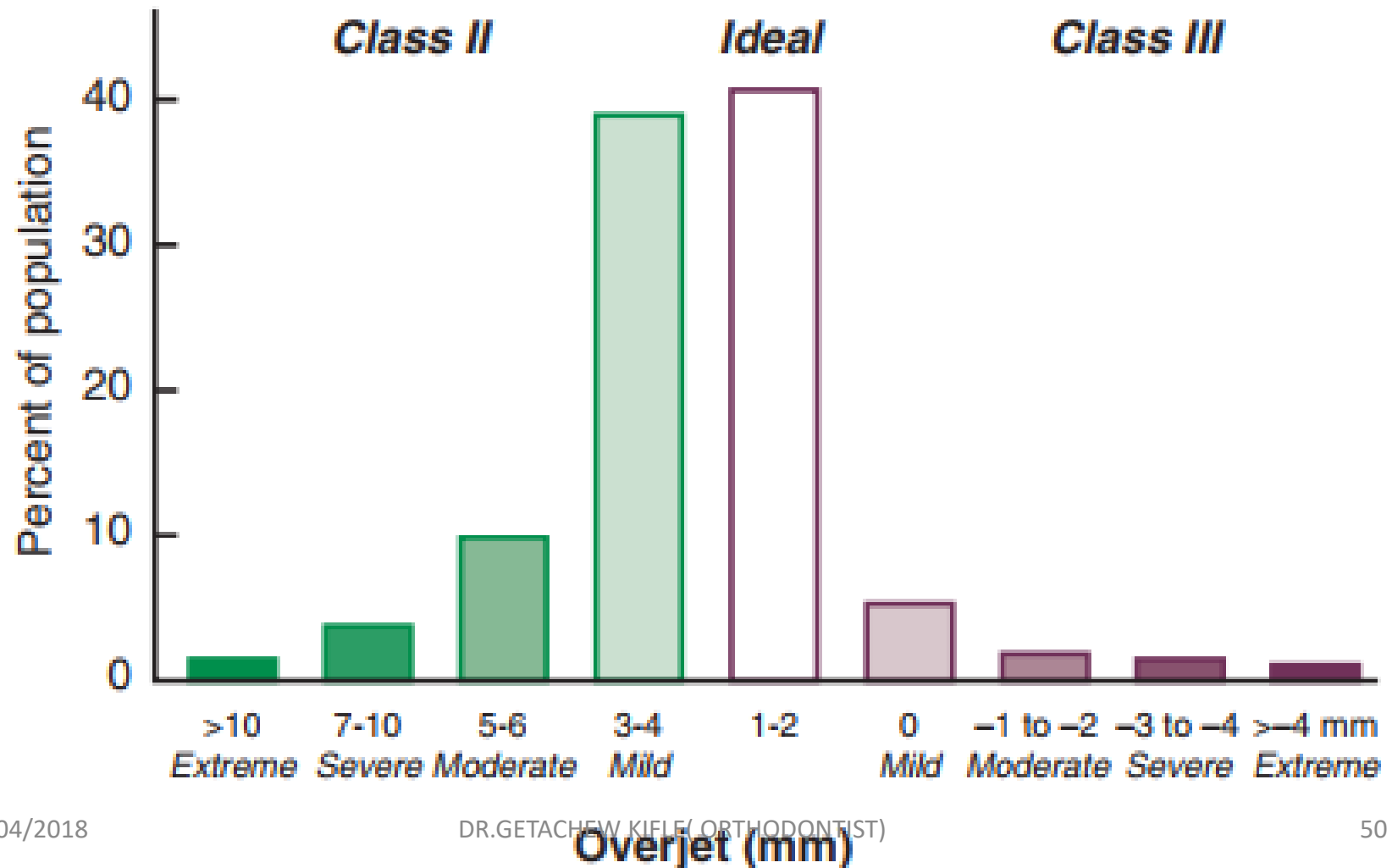


Ideal case for tongue blade therapy



Tongue blade used to treat developing anterior cross bite

Degree of severity of Cross- bite

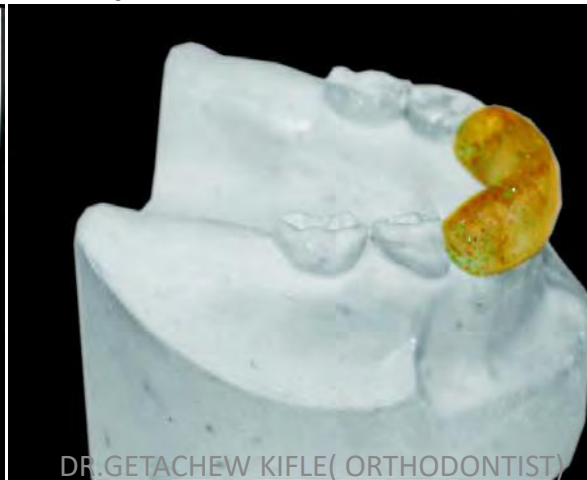


CATALANS APPLIANCE OR LOWER ANTERIOR INCLINED PLANE

- Catlan's appliance basically consists of an inclined plane cemented on the mandibular incisors.
- The name Catlan's appliance is generally associated with appliances which are cemented, hence, not removable in nature.
- The lower inclined plane is constructed at an angle of 45° to the maxillary occlusal plane.



23/04/2018



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51

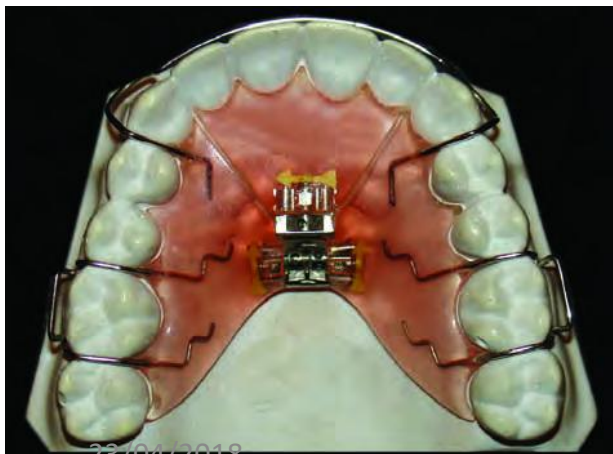
DOUBLE CANTILEVER SPRING/‘Z’ SPRING

- The double cantilever spring or the ‘Z’ spring, as it is more frequently called, is one of the most frequently used appliance to correct anterior tooth/ teeth cross bites.
- The spring consists of a double helix between two parallel arms and the inferior arm extends as the retentive component in the acrylic base plate.
- The parallel arms can be activated as per the requirement to either push the entire tooth labially or just the mesial or distal aspect of the tooth to correct a mesio-palatal/lingual or disto-palatal/lingual rotation of the tooth respectively.



Face Mask or Face Mask Along with RME

- In cases of anterior cross bite due to an actual skeletal deficiency of the maxilla, at times it is possible to mesialize the maxilla using a protraction facemask(reverse head gear).
- If the maxilla is narrow a rapid maxillary expansion screw may be employed simultaneously to aid in the transverse expansion of the maxilla.



23/04/2018



DR.GETACHEW KIFLE(ORTHODONTIST)

53

2.CORRECTION OF ANTERIOR CROSS BITES IN ADOLESCENTS AND ADULTS

✓ **Screw Appliances**

- Mini or medium screws may be used to correct single tooth or segmental anterior cross bites in adults.
- Adequate amount of space is essential to achieve correction otherwise the results will be compromised



✓ Fixed Appliances

- Fixed appliances can be used to correct single tooth or multiple tooth of segmental anterior tooth/teeth cross bites at practically any age.
- The appliance therapy may or may not be accompanied by the use of extractions to create space



CORRECTION OF POSTERIOR CROSS BITES

➤ Screw Appliances

- The various types of screws can be used to correct single tooth or segmental posterior tooth cross bites in patients of all age groups.
- The patient has to be cooperative enough to maintain the appliance and activate the screw or at least get it activated at regular intervals.

➤ Coffin Spring

- This omega shaped wire appliance is capable of correcting cross bites in the young developing dentition.
- The appliance is removable and usually well tolerated by the patients of this age group.
- The expansion produced is slow, and bilaterally symmetrical.
- When used in the mixed dentition stage and with better retention than the usually used Adam's clasps, the appliance is capable of producing skeletal changes.



➤ Quad Helix Appliance

- The quad helix evolved from the coffin spring and overcomes the short comings of the former appliance.
- It is a fixed appliance, soldered to molar bands cemented generally on the first permanent maxillary molars. Reactivation using the three pong pliers
- The forces generated can be increased or decreased depending upon the amount of activation.
- It is a versatile appliance and can be used along with the usual fixed appliance therapy.
- The appliance can produce slow expansion in adolescent and adult patients and skeletal effects in the preadolescents.



➤ The RME Appliance

- The rapid maxillary expansion (RME) involves a hyrax screw type of appliance which produces high forces capable of splitting the mid-palatine suture and bringing about skeletal changes within a matter of days (0.2-0.5 mm/day).
- The RME screw can be incorporated in two type of appliances—one, the banded RME, and the second kind, the cemented RME.



Fixed Orthodontic Appliances

- Fixed orthodontic appliance can be used for correction of posterior cross bites.
- With expanded or constricted depending upon the movement required.
- With auxiliary springs(especially for single tooth x-bite)
- With RME (for severe bilateral posterior x-bite)



➤ Cross-elastics

- *can be used to bring about correction* of individual tooth cross bites in the posterior segment.
- Fixed orthodontic appliances are ideal for the accurate placement of teeth in a dental arch as they provide a three dimensional control over the tooth.

